Housing Price Prediction Using Machine Learning

An Optimal House Price Prediction Algorithm: XGBoost

Hemalata Sharma, Hitesh Harsora, Bayode Ogunleye

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Problem Statement & Dataset

The Problem

Accurately predicting residential property prices is crucial for buyers, sellers, and investors. Traditional methods often fail to capture complex market dynamics and non-linear relationships between features.

Dataset

- 2,930 residential properties
- 79 features
- Dataset: Ames Housing
- Available at: https://www.kaggle.com/datasets/shashanknecrothapa/ames-housing-dataset

Why This Matters

Real-world Impact: Better predictions help homebuyers make informed decisions and reduce financial risk

Technical Challenge: Housing prices involve non-linear relationships, missing data, and multicollinearity

Learning Goals: Master feature engineering, model comparison, ensemble methods, and hyperparameter tuning

Related Works: Primary Paper

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Key Contributions:

- Provides a comparative evaluation of multiple ML regression models for house price prediction.
- Demonstrates that XGBoost achieves superior predictive performance over other models.
- Identifies key influential housing features through feature importance analysis.
- Provides interpretable feature importance analysis

Survey on Related Work

House Price Prediction Using Machine Learning: A Survey of Literature

Nor Hamizah Zulkifley, Shuzlina Abdul Rahman, Nor Hasbiah Ubaidullah, Ismail Ibrahim

2020 • International Journal of Modern Education and Computer Science

- Highlighted importance of feature engineering
- Identified boosting and ensemble methods as high-performing
- Reviewed ML models used in housing price prediction

DOI: https://doi.org/10.5815/ijmecs.2020.06.04

House Prices Prediction with Machine Learning Algorithms

Chenchen Fan, Zechen Cui, Xiaofeng Zhong

2018 • Proceedings of the ACM International Conference

Applies multiple machine learning regression models to the Ames Housing dataset and compares their predictive accuracy, highlighting the effectiveness of ensemble-based approaches.

DOI:https://dl.acm.org/doi/epdf/10.1145/3195106.3195133

Optimizing Polynomial and Regularization Techniques for Enhanced Housing Price Prediction Accuracy

Preethi, D.H.R.Murthy, Vani Hiremani, Raghavendra M.Devadas, R.SAPNA

2025 • SN Computer Science (Springer)

Investigates polynomial regression and regularization techniques (Ridge/Lasso) for housing price prediction and demonstrates how proper model tuning improves generalization performance.

DOI:https://doi.org/10.1007/s42979-024-03578-7

How Exactly Does the Selected Paper Solve the Problem?

Problem Formulation

The paper treats house price prediction as a supervised regression problem where the goal is to map housing features to a continuous target variable (SalePrice).

Core Approach

The authors preprocess the data (handle missing values, encode categorical variables, scale features) and implement multiple regression models for comparison.

Main Algorithm: XGBoost

XGBoost builds an ensemble of decision trees sequentially.
Each new tree learns from the residual errors of the previous trees.
This iterative boosting reduces prediction error step-by-step.

Evaluation

Performance is measured using RMSE, MAE, and R^2 to identify the best-performing model.

Why Does It Make Sense?

- Housing Prices Are Nonlinear

House prices depend on complex interactions (location × size × quality). Linear models struggle to capture this.

- XGBoost Captures Feature Interactions

Tree-based boosting automatically models nonlinear relationships and interactions without manual feature engineering.

- Built-in Regularization

XGBoost includes regularization parameters to prevent overfitting, making it suitable for structured tabular datasets.

- Proven Performance

Boosting algorithms consistently outperform single models in structured data problems.

Methodology and High-Level Solution

1 Data Preprocessing → **2** Feature Engineering | **3** Model Training | **4** Evaluation & Analysis

Key Techniques

Data Preprocessing: Handle missing values using median/mode imputation; remove outliers beyond 3 standard deviations; log-transform target variable (SalePrice)

Feature Engineering: Create interaction features (e.g., TotalSF = 1stFlrSF + 2ndFlrSF); encode categorical variables using target encoding; polynomial features for non-linear relationships

Models: XGBoost (primary), Random Forest, Gradient Boosting; 5-fold cross-validation; GridSearchCV for hyperparameter optimization

Evaluation: RMSE on log-transformed prices; R^2 score; feature importance analysis using SHAP values

Project Timeline & Work Plan

Duration	Deliverables
1 weeks	Annotated bibliography Methodology summary Clean dataset EDA report with visualizations
2 weeks	Working code repository Trained models
2 weeks	Performance comparison report Feature importance plots
1 weeks	Final report Presentation deck GitHub repository

Thank You